📊 Final Report: Food Delivery Time Prediction

1. Introduction

The objective of this project was to develop a machine learning model to predict whether a food delivery would be **Fast** or **Delayed**. This prediction is crucial for improving delivery efficiency, optimizing logistics, and enhancing customer satisfaction.

The problem was framed as a **binary classification task**, using features such as distance, weather conditions, traffic levels, and delivery-related variables.

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2. Data Preprocessing & Feature Engineering

Data Cleaning

- Missing values were handled using forward fill and statistical imputation.
- Dataset consistency was ensured by removing invalid or incomplete entries.

Feature Engineering

- The **Haversine formula** was used to calculate the geographical distance between restaurant and customer locations.
- A binary target variable (**Delivery_Status**) was created:
  - 0 → Fast Delivery
  - 1 → Delayed Delivery

Feature Transformation

- Categorical variables (weather, traffic, vehicle type) were encoded using **Label Encoding**.
- Numerical features were normalized using **StandardScaler** to improve model performance.

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3. Model Implementation

Three machine learning models were implemented:

1. Naive Bayes

- A probabilistic model assuming independence between features.
- Fast and efficient but limited in handling complex relationships.

2. K-Nearest Neighbors (KNN)

- Classifies based on similarity with neighboring data points.
- Hyperparameter tuning was used to find the optimal number of neighbors.

3. Decision Tree

- A rule-based model that splits data based on feature importance.
- Hyperparameters like max_depth and min_samples_split were tuned to prevent overfitting.

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4. Model Evaluation

The models were evaluated using:

- **Accuracy**
- **Precision**
- **Recall**
- **F1-score**
- **Confusion Matrix & ROC Curve**

Performance Summary

Model	Accuracy	Precision	Recall	F1-Score
Naive Bayes	~100%	1	1	1
KNN	~55%	0.57	.52	.55
Decision Tree	~97%	1	0.97	0.97

(Values may vary depending on dataset and tuning)

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5. Key Findings

- **Distance** is the most influential factor affecting delivery delays.

- **Traffic conditions** significantly impact delivery time, especially during peak hours.
- **Weather conditions** (rain, storms) increase the likelihood of delays.
- **Decision Tree model consistently outperformed other models** across all metrics.
- **Naive Bayes struggled** due to unrealistic independence assumptions.

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6. Model Comparison Insights

- **Naive Bayes**
  - Fast but less accurate
  - Not suitable for complex feature interactions
- **KNN**
  - Good performance but computationally expensive
  - Sensitive to feature scaling and choice of K
- **Decision Tree**
  - Best overall performance
  - Easy to interpret and implement
  - Handles non-linear relationships effectively

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7. Actionable Recommendations

For Delivery Operations

- Use predictive models to **identify potential delays in advance**.
- Optimize routes using **real-time traffic data**.
- Assign deliveries based on **vehicle efficiency and distance**.

For Business Strategy

- Provide **accurate delivery time estimates** to customers.
- Offer **priority delivery options** in high-delay zones.
- Improve customer experience by proactively managing delays.

For System Improvement

- Deploy **Decision Tree model** in real-time systems.
- Integrate external APIs for:
  - Traffic updates
  - Weather conditions
- Continuously retrain models with new data for better accuracy.

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8. Conclusion

This project demonstrates how machine learning can significantly enhance food delivery systems by predicting delays accurately. Among all models, the **Decision Tree classifier emerged as the best-performing model**, offering both high accuracy and interpretability.

Implementing such predictive systems can lead to **improved operational efficiency, reduced delivery delays, and enhanced customer satisfaction**.

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9. Future Scope

- Use advanced models like **Random Forest or XGBoost** for improved performance.
- Incorporate **real-time GPS tracking data**.
- Build a **live prediction dashboard** for delivery monitoring.

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